

Qatar-2b

R-0971

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180610 · created 2026-07-29T17:38:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458279.7813 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.221 +/- 0.027
Transit depth (Rp/Rs)^2	4.89 +/- 1.21 [%]
Orbital Inclination (inc)	89.93 +/- 1.46
Airmass coefficient 1 (a1)	1.352 +/- 0.078
Airmass coefficient 2 (a2)	-0.1345 +/- 0.0092
Scatter in the residuals of the lightcurve fit is	5.7 %
Best Comparison Star	None
Optimal Aperture	8.58
Optimal Annulus	12.16
Transit Duration (day)	0.0803 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.221 $\pm$ 0.027 vs 0.16327 (deviation 1.67 σ)
Duration fit vs published	0.0803 d vs 0.0767 d (deviation 1.35 σ)
Mid-time O-C	1.1 min
Depth SNR	6.4
Residual scatter	5.7 %

Light curve

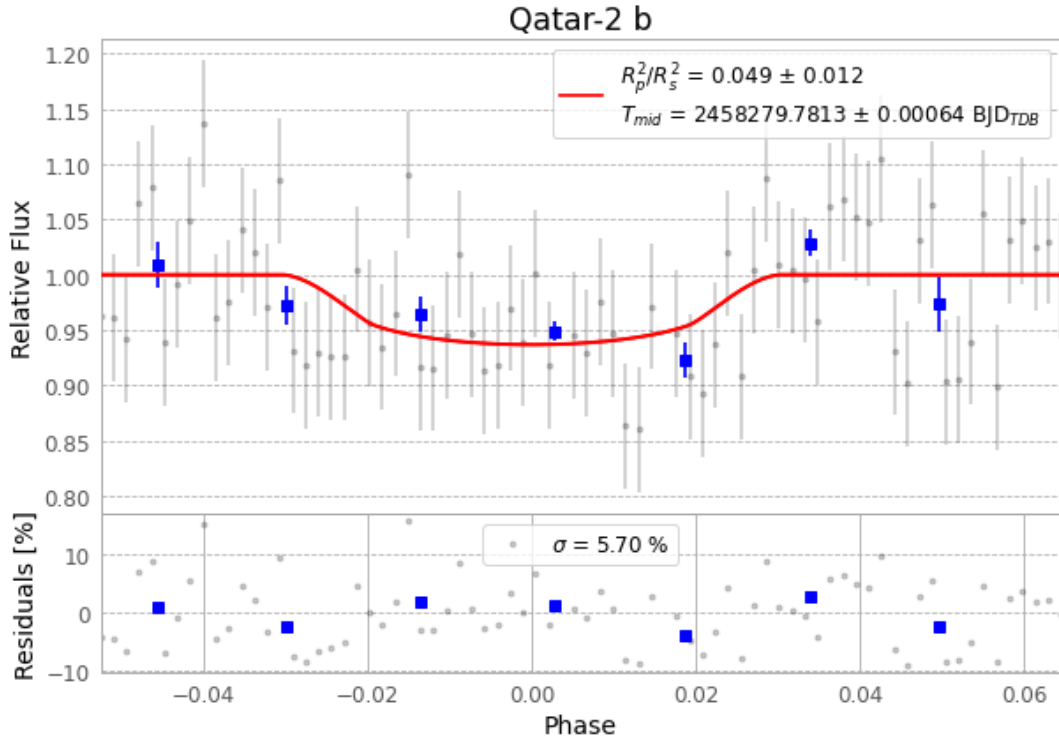


Figure 1 — FinalLightCurve_Qatar-2 b_2018-06-09.png (SHA-256 adae70b7cd44c825...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 461], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b66dfd329e3c75fc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

76 FITS frames (50 MB), Qatar-2180610045712.FITS → Qatar-2180610084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180610.log (10c617f8b878...), FinalLightCurve_Qatar-2 b_2018-06-09.png (adae70b7cd44...), FinalLightCurve_Qatar-2 b_2018-06-09.csv (a0c1868840f9...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 17:38Z.